HS-25—all have to perfect. We had got Indian public sector units like MIDHANI and some private players to work on materials like KC-20WN. But these would not be used for the first few flights; we used material procured from the original suppliers of SEP and got some of the systems, like the combustion chamber, here.

Next, we had to get the tankages for the fuel and oxidiser. We approached Air Liquide, an American company that supplied tankages to SEP. Air Liquide, France, made four of them for us, and they were to be shipped two each per flight. Each flight cost Rs 30 lakh, which put the transportation cost at Rs 60 lakh. When I made an inspection at the factory only one tankage was fabricated, three remained to be made. It raised many eyebrows when I asked the only tankage ready, to be flown down immediately.

To many, it made no sense as it meant there would be at least three trips as a flight cannot carry more than two, and this would send the transportation cost up to Rs 90 lakh. I had my reasons, but this was not the time to explain. So, I got the first one immediately, the second and third in another flight, and the last in a third flight. It did cost Rs 30 lakh more, and the comptroller and auditor general did make an issue out of this.

I was not much bothered about these bean counters, for I had explained the reason to the ISRO chairman and the few people who mattered. Dhawan explained the matter to the Indian Prime Minister and got the additional expenditure ratified. Now, here was the reason. I feared an embargo due to American pressure.

Not many know I had just beaten an embargo while getting the gas bottles from another American company. The bottles, which hold the gas that gives the initial pressure to move the fuel and oxidiser into the combustion chamber, are made of a titanium alloy. The difficult part of making the bottles is machining of the alloy from a thickness of 12 mm to the required 1.5 mm-2 mm thickness.

The US manufacturer was to ship sixteen hemispheres, which we were to be put together into eight gas bottles. Just when we were awaiting the cargo at Madras port, we got a call from the manufacturer saying he had